

# Quiz 2 Answer Key

Wednesday, March 4, 2026 4:46 PM

## Question 1:

In a client-server architecture

- ☐ Clients can communicate directly with other clients.
- ☐ Clients always initiate communication
- ☐ All hosts act as both clients and servers simultaneously
- ☐ Servers only communicate with other servers

## Question 2:

SMTP allows email servers to exchange emails safely through native authentication and encryption mechanisms.

- ☐ True
- ☐ False

## Question 3:

Which transport protocol is most commonly used by HTTP?

- ☐ UDP
- ☐ TCP
- ☐ QUIC

## Question 4:

HTTP is stateless, i.e., each HTTP request-response pair is handled independently.

- ☐ True
- ☐ False

## Question 5:

In socket programming with UDP, a new socket is created for every client-server process pair.

- ☐ True
- ☐ False

## Question 6:

In socket programming with TCP, a new socket is created for every client-server process pair.

- ☐ True
- ☐ False

## Question 7:

Which transport protocol is most commonly used by DNS queries?

- ☐ UDP
- ☐ QUIC
- ☐ TCP

## Question 8:

A user enters [www.nytimes.com](http://www.nytimes.com) into a browser. List the major protocols and steps required to load that webpage. Assume that the IP address of nytimes.com is unknown.

**Answer:** User enters URL into the browser. DNS lookup, looks up corresponding domain name into the local cache or OS resolver. DNS returns the IP address. Browser opens a TCP connection to the server. HTTPS security through TLS handshake. Browser sends HTTP request. Gets HTTP back and renders to user.